First class – 01.

08/07/2026
Wednesday.

1) CPU → Brain of a computer

2) Memory Device → RAM, ROM, HDD (Storage)

3) Input (keyboard, mouse, etc)

4) output (Monitor)

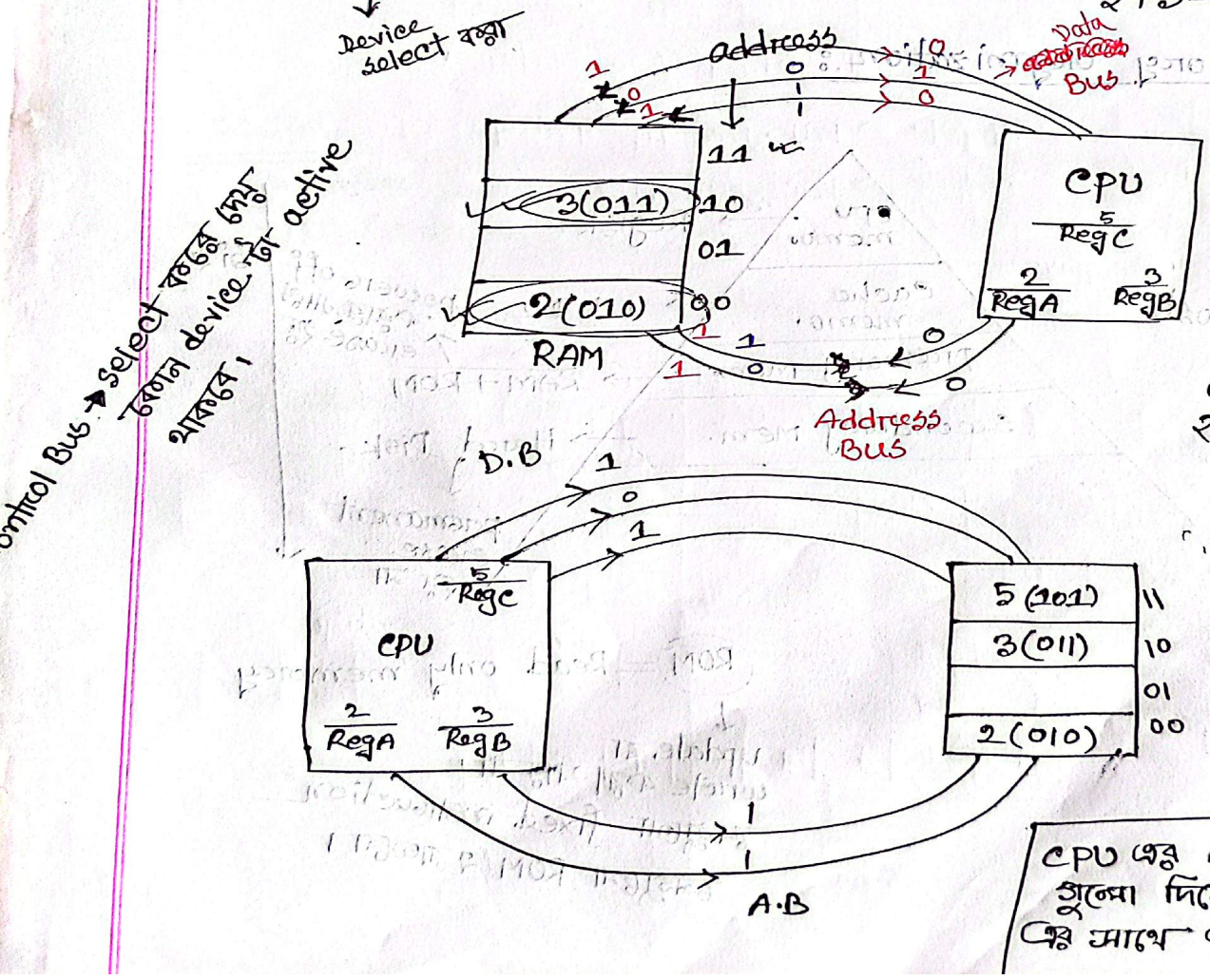
computer কে যেকোনো ধরনের কাজ করতে হলে তার সব data RAM এ থাকতে হবে।

5) Bus ⇒ wire (use for connection)

- Address Bus.
- Data Bus.
- Control Bus.
  - ↓ Device select করা

Address Bus এর direction হবে CPU থেকে device গুলোর দিকে। Not device to CPU ☺

2+3 = 5 (11)



CPU যেই location কে access করতে চায় তেমন address Bus এ পাঠাবে।

Bus address পাঠানো হল CPU থেকে।

D.B ⇒ bi directional.

A.B ⇒ uni directional.

CPU এর A.B এর wire/D.B গুলো দিয়েই প্রত্যেক device এর সাথে connected.

common wire

Keyboard, printers → IO device.

MWTC → Memory write command.

MRDC → Memory Read command.

IOWC → Input output write command.

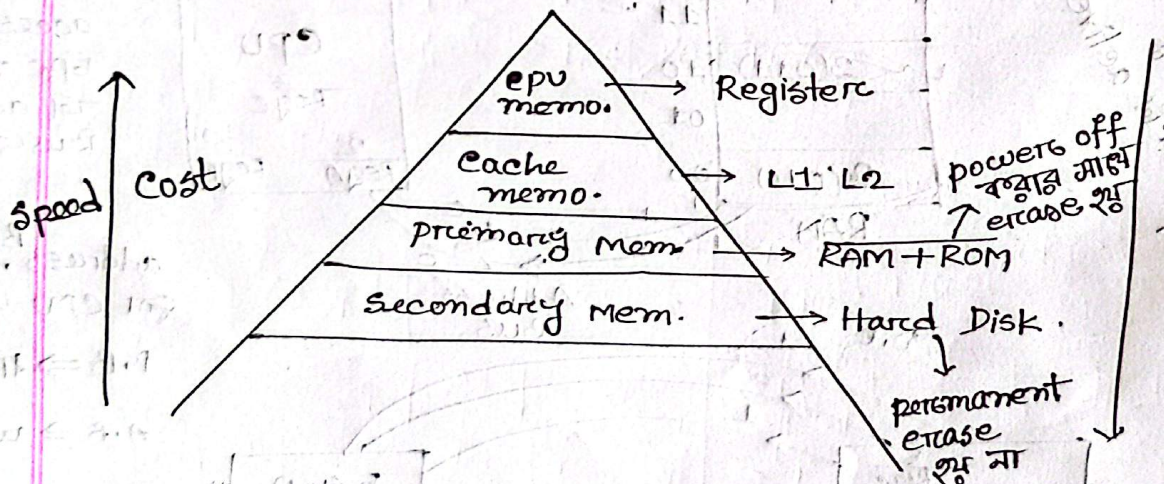
IORC → " " Read "

Computers read → keyboard

computers write → printers

↳ (input নিচ্ছি)

Memory organization :



ROM = Read only memory

↓

update বা write করা যায় না,

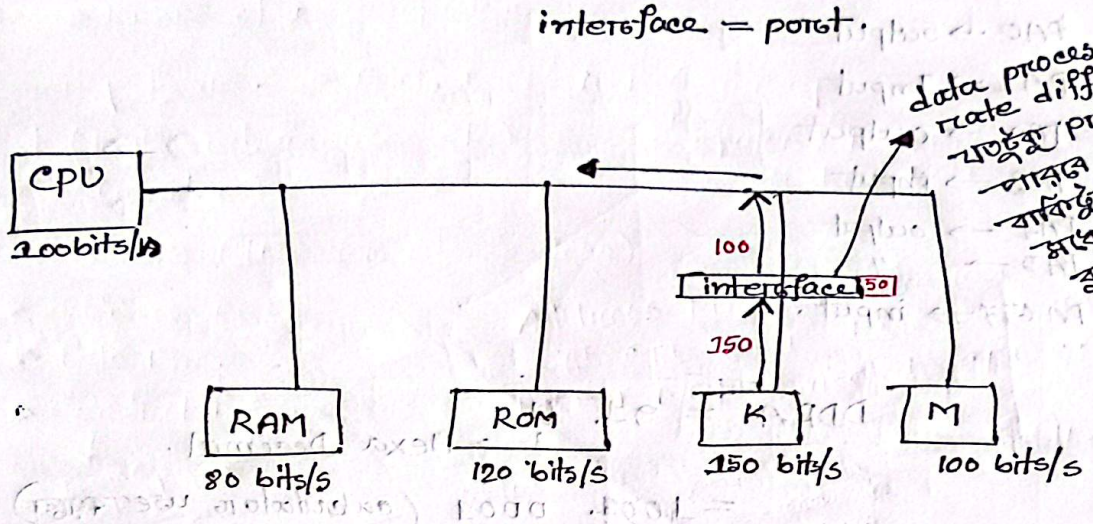
এগুলো fixed instruction

সেগুলো ROM এ রাখবো।

Lecture – 02.

12/07/2026.
Sunday

interface = port.



8086

↓ Interface port A, port B, port C, port D.

82C55. (4 টা port আর 8 টা pin)

There are three ways of transferring data between a micro-computer and physical I/O devices:

i) programmed I/O

→ pin configuration এর উপর base করে-

programmed I/O → Individual I/O → use হয় DDR
প্রত্যেক pin কে আলাদা আলাদা ভাবে config করতে পারবো।

→ parallel I/O
↓
পুরো port কে config করতে হবে।

PA = port A = 8 bit

port A → DDRA → registers
Direction
Data

DDRA
↓
port A

DDR → 8 bit

PA7							PA0
0	0	1	1	0	1	0	1

PA1
LSB

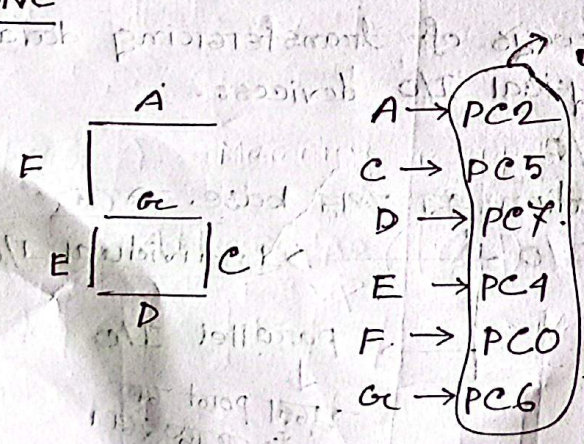
0 → input
1 → output

PA0 → output
PA1 → Input
PA2 → output
PA3 → input
PA4 → output
PA5 → "
PA 6, 7 → input.

DDRA = 91 H
↳ Hexa Decimal.
= 1001 0001 (calculator use করবো)

Input → PA1. PA2 PA3 PA5 PA6.
output → PA0 PA4 PA7.

Question solve



DDRC =

1	1	1	1	0	1	0	1

Based on Memory mapping :

programmed I/O

memory mapping

Standard I/O

↓

memory and I/O information আলাদা থাকে

Memory mapped I/O

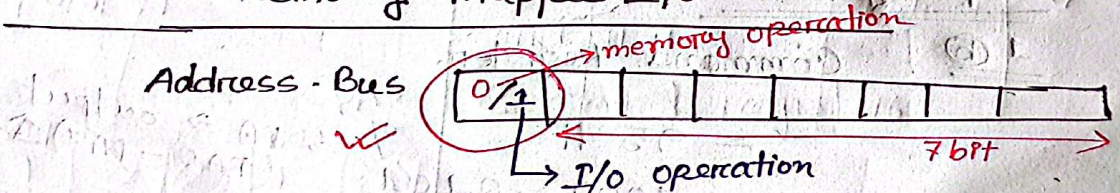
↙

প্রত্যেক location আলাদা থাকে।

M / $\overline{IO}$ = 1 (Memory)
= 0 (I/O).

কোন operation নিম্নে করতে করবে তা decide করে

Memory mapped I/O



A.B = 8 bit
$= 2^8 = 256$ memory location

7 bit = $2^7 = 128$

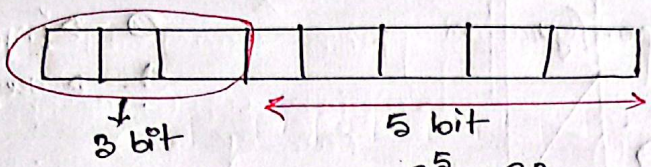
256 memory access করার কথা ছিল কিন্তু করছি 128.

∴ কাজ কমে 50 %

Fall

1(b)

A.B = 8 bit = 2^8 = 256.



$= 2^5 = 32$

$\therefore 256 - 32 = \boxed{224}$

$\frac{224}{256} \times 100\% = 87.5\%$

✓ Standard I/O → পুরো memory utilize
→ memory and I/O diff তাই দুটোর access করার জন্য আলাদা command লাগবে।

✓ memory mapped I/O → memory utilize reduce হয়।
→ same command দিয়ে access করা যায়।

First half → Bust $600 + \frac{2 \times 10^9}{50 \times 1024} + 500 = 80141900$

Rest half → cycle $\left(600 + \frac{16 \times 10^9}{50 \times 1024} + 500\right) \times \frac{2}{16} = 80141900$

$\therefore$ Hence proved

Lecture - 03 (online class)

Interface এর দরকার কি?

⇒ processor rate বলল কিন্তু keyboard or mouse থেকে sending rate বেশি তাহলে balance করার জন্য interface দরকার। interface দেওয়ার processor যতটুকু process করতে পারবে তখন তাকে ততটুকুই send করা হবে। বাকিটুকু store করে রাখবে।

parallel I/O তে Input বা output আকারে configure করতে হবে পুরা port কে। Individual ভাবে একটা pin কে input/output আকারে config. করতে পারবো না। Input বানালে পুরা port ই input হবে, output বানালে পুরা port-ই output হবে।

2| Interrupt driven I/O :-

processor normally নিজের কাজ করতে থাকবে। যখন I/O device থেকে কোনো interrupt আসবে তখন সে আগের কাজকে pause করবে। interrupt এর জন্য যে কাজটা করা দরকার করবে + করা শেষ হলে আবার normal execusion এ ফেরত যাবে।

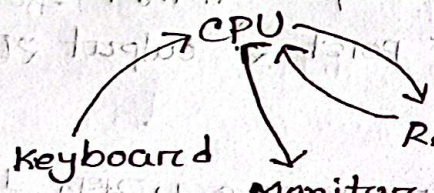
→ External interrupts : I/O device থেকে আসবে। অর্থাৎ keyboard, mouse etc

→ Internal " : processor এর ভেতর থেকে আসবে। নিজেই নিজের কাজে বাধা দিবে।

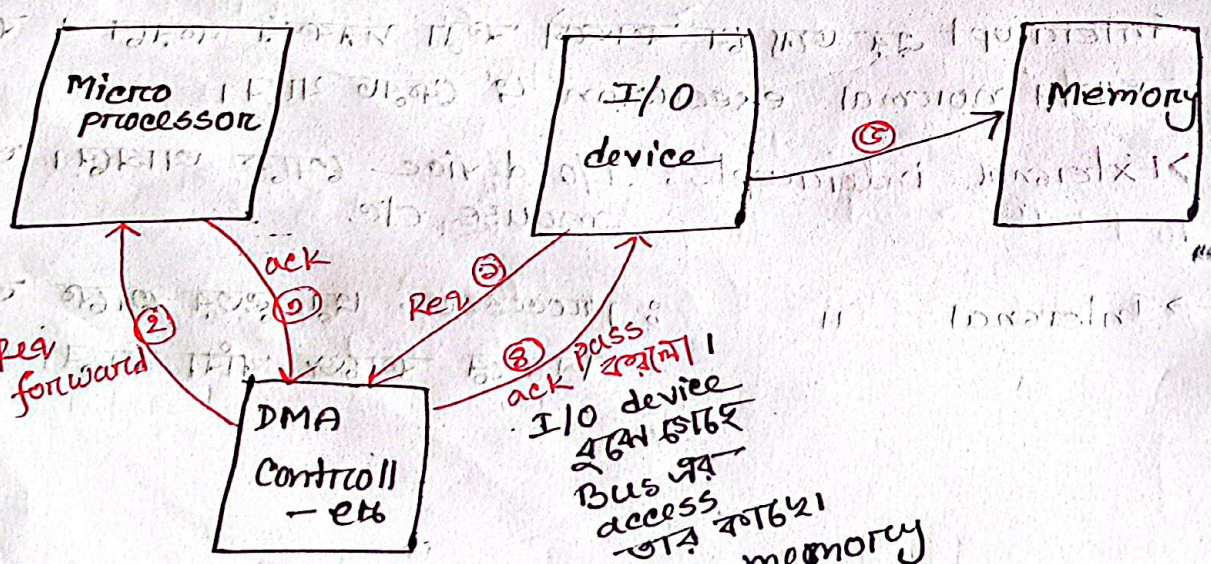
External interrupt

- Maskable
  - which user can control (on/off)
- Non-maskable interrupts
  - which user cannot control (always on)

③ Direct memory Access (DMA) : এর topic এ math আছে।



⟹ এটা না করে I/O device এর সাথে RAM এর direct connection বা communication ~~করা~~ stablished করাটাই DMA।



DMA → i) Bust mode.
ii) cycle strealing mode.
iii) Transparent mode.

(*) **Bust Mode :-**

Step 1 : Take Bus access (microprocessor থেকে access নিবে)
Step 2 : Data transfer
Step 3 : Give Back Bus access (micro processor থেকে access ফেরত দেবো)

∴ Total time = take Bus access + Data transfer + Give back Bus access.

Math Transfer of bus control in either direction, from processor to device or vice-versa, takes 250 ns. I/O device has data transfer rate of 75KB/sec.

(a) Suppose we employ DMA in a bust mode. How long does it take to transfer a block of 256 bytes?

Soln :

(a) Total time = take bus access + Data transfer + Give back bus access

$$= 250 \text{ ns} + \frac{256 \times 10^9}{75 \times 1024} \text{ ns} + 250 \text{ ns}$$

$$= 3.33 \times 10^6 \text{ ns}.$$

75 KB/s
1S 75KB
= 75×1024 B

∴ 75×1024 B is transferred in 1S

∴ 1B " " " $\frac{1}{75 \times 1024}$ S

∴ 256 B " " " $\boxed{\frac{256}{75 \times 1024}}$ S.

(*) cycle stealing mode:

Repeat until whole data is transferred

step 1 : Take Bus access.
step 2 : Data transfer (a fixed portion of data)
step 3 : Give back Bus access.

Total time = (Take Bus access + Data transfer (fixed portion) + Give Back Bus access) × no. of cycles

$= \frac{\text{total Data}}{\text{Data transferred per cycle}}$.

Lecture - 01.
19 July 2026.
(Sunday).

Burst mode

Total time = Take Bus Access + Data Transfer + Give Back Bus Access.

C.S mode

Total time = (Take Bus Access + Data Transfer (fixed) + Give Back Bus Access) × No. of cycle.

$$\text{No. of cycle} = \frac{\text{total Data}}{\text{Data transferred per cycle.}}$$

practice problems.

(1) (a)

$$\text{Total time} = 250\,ns + \frac{512}{100\times1024}\,s + 250\,ns$$

$$= 250\,ns + \frac{512\times10^9}{100\times1024}\,ns + 250\,ns$$

= (Ans)

$1\,KB = 2^{10}\,B$

$1\,MB = 2^{20}\,B$

$1\,GB = 2^{30}\,B$

100 KB/s ⇒ 1s. . . . 100KB

$= 100\times1024\,B$

$100\times1024\,B$ - - - - 1s

$\therefore 1\,B$ - - - $\frac{1}{100\times1024}$

$\therefore 512\,B$ - - - $\frac{512}{100\times1024}\,s$

⑥

Total time $= \left(250\,ns + \frac{1 \times 10^9}{100 \times 1024} + 250\,ns\right) \times \frac{512}{1}$

↳ data transfer rate

$=$

2।

(a) Total time $= 200\,ns + \frac{128 \times 10^9}{50 \times 1024}\,ns + 150\,ns$

$=$

(b) Total time $= \left(200 + \frac{1 \times 10^9}{50 \times 1024} + 150\right) \times \frac{228}{1}$

$=$

3। (b) total time $= \left(100 + \frac{2 \times 10^9}{20 \times 1024} + 150\right) \times \frac{256}{2}$

$=$

4। (b) total time $= \left(300 + \frac{4 \times 10^9}{40 \times 1024} + 200\right) \times \frac{512}{4}$

$=$

5। (a) total time বেশী যেটা হবে, Bus access দেয়া ও নেয়ার ফ্রিকোয়েন্সি।

$\Rightarrow 10000250\,ns = 100\,ns + \frac{512 \times 10^9}{x \times 1024} + 150\,ns$

x → KB

1024 → B

$$\Rightarrow 10000250 = \frac{100 \times 1024x + (512 \times 10^9) + 150 \times 1024x}{1024 \times x}$$ let,

Data transfer rate $= x$ KB/S

$\Rightarrow x = 50$ KB/S.

(b) Data transfer rate $= 50$ KB/S

256 B In Burst mode,

$$\text{time}_1 = 100\text{ ns} + \frac{256 \times 10^9}{50 \times 1024} + 150\text{ns}$$

256 B In cycle stealing mode,

$$t_2 = \left(100\text{ ns} + \frac{12 \times 10^9}{50 \times 1024} + 150\text{ns}\right) \times \frac{256}{2}$$

$\therefore$ Total time $= t_1 + t_2$

$=$

Spring 26.

1. (a)

Shafi,

$$100ns + \frac{1021 \times 10^9}{\left(\frac{x}{10^6} \times 1021\right)} ns + 200ns = y \longrightarrow (i)$$

x kB/μs

1 μs - - - - x kB

$\therefore 10^{-6}$ s - - - - x kB

$\therefore$ 1 s $\quad \frac{x}{10^{-6}}$ kB

$= \frac{x}{10^{-6}}$ kB/s

Rafi,

$$100ns + \frac{1021 \times 10^9}{\left(\frac{2x}{10^6} \times 1021\right)} ns + 200ns = y - 50 \longrightarrow (ii)$$

(i) − (ii) $\Rightarrow$

$$\frac{10^3}{x} - \frac{10^3}{2x} = 50$$

$$\Rightarrow x = 10$$

Fall 25 ***

(1) (a) $3 + 5 + 3 + 5 + \dots = 600$

let,

num. of cycles = n

75 ← cycles $3B \rightarrow \frac{n}{2}$ cycle $\Rightarrow$ total Data $= 3 \times \frac{n}{2}$

75 ← cycles $5B \rightarrow \frac{n}{2} \Rightarrow$ total Data $= 5 \times \frac{n}{2}$

$$\Rightarrow \frac{3n}{2} + \frac{5n}{2} = 600$$

$$\Rightarrow n = 150$$

24 B

3 + 5 + 3 + 5 + 3 + 5

6

প্রত্যেক cycle = 2 B

" = 5 B

$\therefore$ total = 2 × 5 B

= 10 B

2B is send per cycle for 5 cycles
$= 2B \times 5 = 10B$

3B for 75 cycles,

$$t_1 = \left(600\ ns + \frac{3 \times 10^9}{40 \times 1024}\ ns + 500\ ns\right) \times 75$$

$=$

5B for 75 cycles,

$$t_2 = \left(600\ ns + \frac{5 \times 10^9}{40 \times 1024}\ ns + 500\ ns\right) \times 75$$

$=$

$\therefore$ total time $= t_1 + t_2$

$=$

Summer 25

11 @ 1440 KB/minute

1 min = 60 s ..... 1440 KB

$\therefore$ 1 s ..... $\frac{1440}{60}$ KB

$= 24$ KB

24 KB/s.

cycle stealing mode,

$$\left(200\ ns + \frac{x \times 10^9}{24 \times 1024} + 300\ ns\right) \times \frac{3648}{x} = 148456500$$

$\Rightarrow x = 96$

বড় থেকে ছোট = গুন

1 micS = 10^3 ns

1 ms = 10^6 ns

Spring 25

1(a) $\left(0.2\times10^3 \text{ ns} + \frac{8\times10^9}{96\times1024} + 0.1\times10^3 \text{ ns}\right) \times \frac{x}{8}$

$= 62960800$ ns

Lecture – 05.

22/07/2026. (Wednesday)

Transparent Mode : Theory আগে।

✓ DMA (Data memory access) এ Transparent Mode efficient.

✓ I/O device and processor parallely কাজ করে তাই এর execusion time কম।

Topic : Processor

8086 PIN

Address Bus = 20 bit

Data " = 16 bit.

AD0 – AD15

Multiplexed Pin

multiple work

AD → Data

↓

Address

Address Bus পাঠানোর সময় Data Bus এর প্রয়োজন হয় না।

আবার Data Bus এর সময় address Bus পাঠানোর দরকার নেই।

তাই একটা pin দিয়ে multiple কাজ করা যায়।

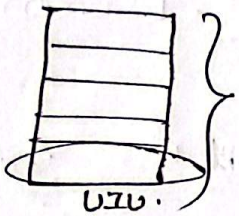
✓ what is multiplexed pin? with example.

1.2 GHz → 1s এ 1.2G যতগুলো execution করতে পারে।

(32)-bit μp → 32 bit Data Bus.

(64)- " " → 64 " " → Better than 32 bit.

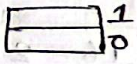
Topic: Memory Size Calculation



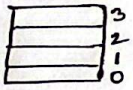
No. of room = 20.
Size of each room = 30 sq. feet.
UIU size = 20×30 sq. feet.

∴ Total size = No. of rooms × size of each room

A.B = 1 bit
= (0, 1)



A.B = 2 bit



A.B = n bit
= 2^n

Ram size = No. of blocks × size of each blocks
= $2^{A.B}$ × Data Bus.

Data Bus যার unit যা আছে সেই unit এর উপর।

no. of blocks নির্ণয় করতে A.B এর উপর

RAM size = $2^{A.B} \times D.B$ **

1km = 1000 m
বড় থেকে ছোট = Multi
ছোট " বড় = Division

(1) A.B = 20 bit
D.B = 8 bit

RAM size = $(2^{20} \times 8)$ bit

= $\frac{2^{20} \times 8}{8}$ byte = $\frac{2^{20} \times 8}{8 \times 20^{20}}$ MB = 1MB

(2)

A.B = 20 bit

D.B = ? bit

Ram size = 1 MB

∵ RAM size = $2^{A.B} \times D.B$

$\Rightarrow 1\ MB = 2^{20} \times D.B$

$\Rightarrow 1\ MB = 2^{20} \times x$ bit

$\Rightarrow 1 \times 2^{20}\ B = 2^{20} \times x$ bit

$\Rightarrow 1 \times 2^{20} \times 8$ bit $= 2^{20} \times x$ bit

$\Rightarrow x =$

(3)

A.B = ?? bit

D.B = 8 bit

RAM size = 1 MB

∵ RAM size = $2^{A.B} \times D.B$

$\Rightarrow 1\ MB = 2^{x} \times 8$ bit

$\Rightarrow 1 \times 2^{20} \times 8 = 2^{x} \times 8$

$\Rightarrow 2^{20} = 2^{x}$

$\Rightarrow x = 20$

$16 = 2^{x}$

$\Rightarrow x = \log_2(16)$

Spring 2026.

31(b)

Let, SRAM size = x MB = 4 MB, D.B = 16 bit A.B ?
DRAM$_1$ " = $2x$ MB = 8 MB, D.B = 32 bit A.B ?
DRAM$_2$ " = $4x$ MB = 16 MB, D.B = 64 bit A.B ?

$\therefore x + 2x + 4x = 28$ MB
$\Rightarrow 7x = 28$ MB
$\Rightarrow x = 4$ MB

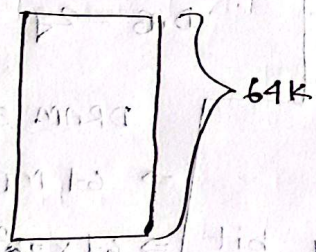
For SRAM :-

SRAM size $= 2^{A.B} \times D.B$
$\Rightarrow 4$ MB $= 2^{A.B} \times 16$ bit
$\Rightarrow 4 \times 2^{20} \times 8 = 2^{A.B} \times 16$
$\Rightarrow \frac{4 \times 2^{20} \times 8}{16} = 2^{A.B}$
$\Rightarrow A.B = 21$ bit

অবজুলোর - ই same ans আসছে।

Fall-25

3(b)



SRAM$_1$

No. of blocks = 64K
$= 64 \times 1K$
$\therefore 2^{A.B} = 64 \times 2^{10}$

64K addressable locations
= No. of Rooms থাকা।
$= 2^{A.B}$

DRAM size $= 2^{A \cdot B} \times D \cdot B$

$\Rightarrow 2 \times$ SRAM size $= 2^{A \cdot B} \times \frac{16}{2}$ bit

$\Rightarrow 2 \times (64 \times 2^{10} \times 16) = 2^{A \cdot B} \times 8$

$\Rightarrow A \cdot B = 18$ bit

SRAM size $= 64 \times 2^{10} \times 16$ bit

$= \frac{64 \times 2^{10} \times 16}{8}$ B

$= \frac{64 \times 2^{10} \times 16}{8 \times 2^{20}}$ MB

$\therefore$ DRAM size $=$ SRAM size এর double.

Summer – 25

3(b) let,

SRAM:

$A \cdot B = x$

$D \cdot B = y$

SRAM size $= 2^{A \cdot B} \times D \cdot B$

$\Rightarrow 8\text{KB} = 2^x \times y$

$\Rightarrow 8 \times 2^{10} \times 8$ bit $= 2^x \times y$ bit

$\Rightarrow 8 \times 2^{10} \times 8 = 2^x \times y \longrightarrow (i)$

DRAM:

$A \cdot B = 2x$

$D \cdot B = 2y$

DRAM size $= 2^{A \cdot B} \times D \cdot B$

$\Rightarrow 64\text{MB} = 2^{2x} \times 2y$

$\Rightarrow 64 \times 2^{20} \times 8 = 2^{2x} \times 2y \longrightarrow (ii)$

CamScanner

(ii) ÷ (i) ⇒

$$\frac{8 \times 2^{20} \times 8}{8 \times 2^{10} \times 8} = \frac{2^{2x} \times 2y}{2^x \times y}$$

$\Rightarrow 2^{13} = 2^x \times 2$

$\Rightarrow x = 12$

(i) $\Rightarrow 8 \times 2^{10} \times 8 = 2^{12} \times y$

$\Rightarrow y =$

Spring - 25

36) RAM Y size = a MB = 9 MB

RAM X " $= \frac{2}{3}a$ MB = 6 MB

∴ RAM X size + RAM Y size = 15 MB

$\Rightarrow \frac{2}{3}a + a = 15$

$\Rightarrow \frac{2a+3a}{3} = 15$

$\Rightarrow 5a = 45$

$\Rightarrow a = 9$ MB

∴ RAM X size $= 2^{A.B} \times D.B$

$\Rightarrow 6\text{ MB} = 2^{22} \times x$

$\Rightarrow 6 \times 2^{20} \times 8 = 2^{22} \times x$

$\Rightarrow x = 12$ bit.

↳ indication

Topic : Flag Register

Result যেটা বের হচ্ছে সেটা কেমন তা Flag বা indication বলে দেয়।

SF = Sign Flag.

SF = 0 (Result positive)

SF = 1 (" Negative)

① Result = 23 H

= 0010 0011

MSB = sign bit

= 0 (+) → SF = 0

= 1 (−) → SF = 1

∴ SF = MSB of the result.

② Result = A2 H

= 1010 0010

SF = 1

Result এর MSB bit টা যা হবে সেটাই SF এ বসাবে।

MSB left এ থাকে

lecture:- 06

26/07/2026..

1)

$$\left(450 + \frac{6\times10^9}{35\times1024} + 350\right) \times \frac{540}{6}$$

=

2) let, total data = x

Burst = $\frac{x}{2}$

cycle = $x/2$

$$\left(600 + \frac{\frac{x}{2}\times10^9}{50\times1024} + 500\right) + \left(600 + \frac{16\times10^9}{50\times1024} + 500\right) \times \frac{\frac{x}{2}}{16} = 80141900$$

ZF (zero flag) :

ZF = 0 (Result Non-zero)

= 1 (Result 0)

Result = 0000 H

@ ZF = 1

CF (carry flag) :

CF = 0 (If no carry)

CF = 1 (If carry).

give num 4bit হলে

2)
$$\begin{array}{r} 1010 \\ +\ 0101 \\ \hline 1111 \end{array}$$

CF = 0

given num = 4bit

result ও = 4 bit

So carry নাই।

2)
$$\begin{array}{r} 1010 \\ +\ 1101 \\ \hline 10111 \end{array}$$

carry CF = 1

given num 32 bit হলে carry থাকা জন্য check result এ 33 num bit.

CF, AF এর জন্য same.

smaller num থেকে large num বিয়োগ করলে carry থাকবে

large num " smaller " " " থাকবে না।

AF (Auxiliary carry) : AF কাজ করে 4 bit নিয়ে। (always).

1100 → AF=0

(+) 1011

(1) 0111 1111

carry so CF = 1.

0101

1010

5

10

4 num bit যোগ করার পর 5 num bit এ carry check করবো। carry আসলে AF=1, না আসলে AF=0

বিয়োগ এর ক্ষেত্রে,

যেহেতু ছোট থেকে বড় বিয়োগ করতেছি তাহলে AF = 1 ;

PF : (Parity Flag) → 8 bit check করবো। LSB থেকে। check করবো কয়টা 1 আছে।

Result = A2B3 H

= 1010 0010 1011 0011

5 টা 1 আছে odd. so PF = 0

even পেলে PF = 1.

OF (over flow)

2 | 1 0

3 | 1 1

5 | 1 0 1

add

case 1

$$\begin{array}{r} + \\ (+)\ + \\ \hline - \end{array}$$

OF = 1

positive দুইটা num যোগ করলে normal scenario তে result + । কিন্তু negative আসা মানে overflow । so OF = 1 ।

case 2

$$\begin{array}{r} - \\ (+)\ - \\ \hline + \end{array}$$

OF = 1

sub.

case 3

$$\begin{array}{r} - \\ (-)\ + \\ \hline + \end{array}$$

OF = 1

$-5 - (+3) = -5-3 = -8$

$-3 - (+5) = -3-5 = -8$

case 4

$$\begin{array}{r} + \\ (-)\ - \\ \hline - \end{array}$$

OF = 1

$+3-(-5) = +3+5 = +8$

এই case গুলা বাদে অন্য case exam এ দিলে OF = 0 দিবো

Question :

$$\begin{array}{r} \text{YF}\ \ H \\ +\ 1\ \ H \\ \hline 80\ \ H \end{array}$$

0111, 1111 → 0000 0001

1000 → 0000 → 1 odd

calculators use করবো

$$\begin{array}{r} 1\ 111 \\ 1111 \\ 0001 \\ \hline 0000 \end{array}$$

result ta carry নেই so, CF = 0.

PF = 0

AF = 1

ZF = 0 (result non zero)

SF = 1

OF = 1

positive আর neg num এর ক্ষেত্রে sign bit দেখতে.

$7FFF = 0111\ 1111\ 1111\ 1111$ (+)

$(+)\ 1 = 0000\ 0000\ 0000\ 0001$ (+)

$8000 = 1000\ 0000\ 0000\ 0000$

OF = 1

positive সংখ্যা যোগ করতে গিয়ে result পেয়েছি neg so overflow. OF = 1.

Lecture – 07.
29/07/2026. (Wednesday)

ADD AX, BX → AX = AX + BX

SUB AX, BX → AX = AX − BX

SUB BX, AX → BX = BX − AX

প্রথম register এর মধ্যে Result টা store হবে।

Ex:

AX = FFFF H

BX = FFFF H

command ADD AX, BX.

Sol^n:

(−) 1111 ← AX = FFFF H (16 bit)

BX = FFFF H + 1111

1 FFFE H

result

result টাও হবে 16 bit.

SF ও

result এর MSB তে যেই value থাকবে তা SF

Result = FFFE

1111 1111 1110

1's ২টা

SF = 1

so, PF = 0

ZF = 0
CF = 1
OF = 0
AF = 1

Ex: 02.

1000
8 bit
0000

(–) AL = 80 h
(–) (+) BL = 80 h
Add AL, BL
100 h
carry
Result = 8 00h
0000

SF = Result এর MSB check
SF = 0
PF = 1 [Result = 0000 0000 ⇒ 0 (⇒ even)
ZF = 1
CF = 1
OF = 1

8 = 1000 (–)
দুইটা neg. num যোগ করে result এর MSB পাচ্ছি (+) ∴ so, OF = 1

Ex: 03

1000
AX = 8000H (–)
(–) BX = 0001H (+)
0000
7FFFH
0111 (+)
1111 1111
8 → even
PF = 1

SF = 0
PF = 1
ZF = 0
CF = 0
OF = 1

CF এর জন্য পুরা num নিবো. [input bit আর result bit same থাকলে carry নেই। আর beshi হলে carry আছে]

AF " " LSB থেকে 4 bit.

Spring 26.

2| (a) (i) ADD AX, BX.

AX = 8A32 H (-)

(+) BX = DFED H (-)

16A1F H

Result = 6A1F H

OF = 1 (-)+(-) = (+)

(+) → OF = 1.

0110 (+)

5 টা 1

0001 1111

odd

PF = 0

(ii) SUB AX, CX

AX = value updated vale বসবে। কারণ ADD করার পর AX = result.

AX = 6A1F H

(-) CX = C321 H

FFFF A6FE H

Result = A6FE H

1111 0001 1110

কোন carry আসেনি তাই AF = 0 অথবা বড় থেকে ছোট বিয়োগ করাতে carry থাকেনি তাই so AF = 0

CF = 1 (ছোট থেকে বড় বিয়োগ)

- অথবা অন্য logic

16 bit এর num যোগ করে 17 num bit এ value পাচ্ছি তাই CF = 1.

Fall 25

(i) ADD BX, AX

BX = CFB1 H

(+) AX = ABCD H

1 7B7E H

result = 7B7E H

carry.

AF = 0

CF = 1

1 1 1 0 (+)

1 0 0 1
1 1 0 1
1 1 1 0

(ii) CX = E412 H

(→) BX = 7B7E H → 0 111 (+)

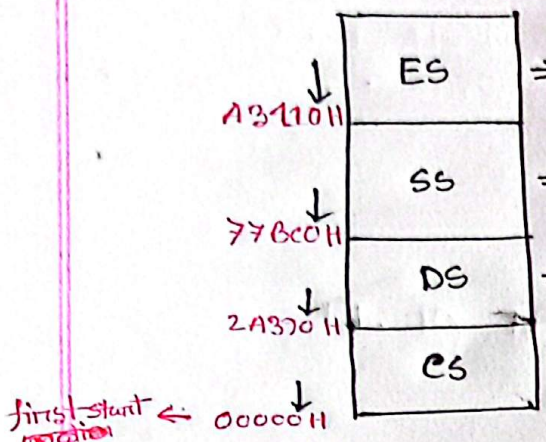
Result = 6894

0 1 1 0 (+)

OF = 1

PF = 0 (3 ते 1 मात्रेचा → odd)

Topic: 8086 Memory segmentations. → এখানে অনেক category তে math



8086 এর Address. Bus = 20 bit (প্রত্যেকটা block এ 20 bit এর)

8086 এ location এর last digit এ zero (0) → fixed থাকে

Register = 16 bit (16 bit এ 20 bit রাখতে পারবো না। So zero কে fixed করে বাকি আগনের 16 bit store করা যায়)

~~Data segment এর~~

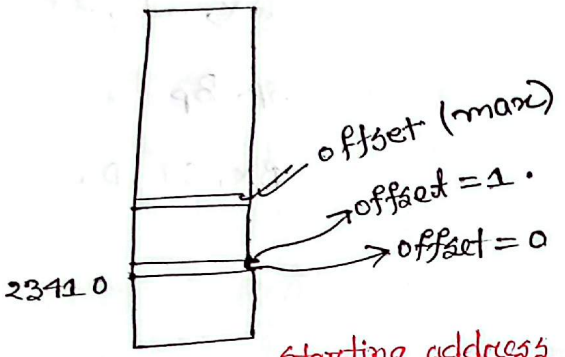
Segment

Segment register.
- CS = 0000 H
- DS = 2A39 H
- SS = 77BC H
- ES = A341 H

Formula:

S. A = Segment register value * 10H
(Starting address)

02/08/2026.
Sunday.



starting address.

8086
Reg size = 16 bit.

1 – bit এর max ⇒ 1
2 – bit " max ⇒ 11
3 – " " max ⇒ 111
16 – " " max = 1111 1111 1111 1111
= FFFF

physical address ⇐ P.A = S.A + offset.

P.A = segment * 10H + offset

offset = P.A – segment * 10H

RAM এর specific memory location

E.A = S.A + offset (max)

E.A = S.A + FFFF H

ending address.

Questions: offset ⇒ 12 bit. → max: 1111 1111 1111 (F F F).

তখন E.A = S.A + FFF H.

offset না বলে দিলে default → 16 bit.

		Segment name	offset
1)	Code segment.	CS	IP
2)	Data "	DS	BX, SI, DI
3)	Stack segment	SS	SP, BP
4)	Extra "	ES	BX, SI, DI

** Important

Question

Code seg.

P.A = 1111 * 10H + 1232

Data seg,

P.A = 3333 * 10H + 0020

SS seg,

P.A = 2526 * 10H + 1100

Logical address:

segment : offset

Example

P.A = 2222 * 10H + 0016

=

✓ offset → sibling থেকে কত দূরে যাবো।

Slide

✓ offset negative আসলে invalid.

Example 3.1.

P.A = 1256Ah,

segment : offset ~~0000h~~ for segments 1256h and 1240h.

Soln:

i) offset = P.A − segment *10 H

= 1256AH − 1256 *10 H

= A H

∴ Logical address = 1256 : 000A → কারণ offset 16 bit এর হবে- কিন্তু পাইছি A মাত্র 4 bit এর।

ii) offset = 1256A − 1240 *10

= 16A H

∴ Logical address ⇒ 1240 : 016A

Ques

offset = 1256A − 2240*10 → আবার দেখেই বলতে পারছি ছোট - থেকে বড় বিয়োগ করতেছি so neg আসবে

= FFFF016A H ⇒ invalid.

→ negative আসা মানে আগে অনেকগুলো F থাকবে।

16 bit এর বেশি হলেই invalid.

CamScanner

Ex 3.2

Physical address = 805D2 H

offset = BFD2 H

∵ Segment = ?

offset → negative হলে invalid

↓

offset এর value 16 bit এর বেশি হলে invalid.

∵ segment * 10H = P.A − offset

⇒ S.A = P.A − offset

⇒ S.A = 805D2 − BFD2 = 74600 H

∵ Segment = $\frac{S.A}{10H}$

= $\frac{74600}{10H}$ = 7460 H.

Ex:

offset = BFD1

P.A = 805D2 H.

= 805D2 − BFD1.

∴ S.A = 74601H → invalid কারণ last digit zero না।

Segment = $\frac{S.A}{10H}$ → invalid. কারণ S.A invalid

= 7460.1

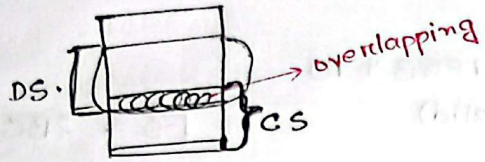
fraction থাকার কথা কিন্তু calculator এ চেক্ করা যাবে না।

Segment বের করার আগে অবশ্যই (PA − offset) বের করবো। বের না করলে incorrect হবে।

** S.A invalid হলেই segment invalid.

তাই first এ S.A বের করবো then check করবো S.A valid নাকি invalid. valid হলে segment বের করবো। S.A invalid হলে segment

Concept :- Overlapping.



overlapping and non-overlapping এর ক্ষেত্রে S.A এবং E.A বের করবো।

Assignment – 7.

Step 1 : Segment গুলোর জন্য S.A এবং E.A বের করা।

Segment 1 :

S.A = 86000 H

E.A = 86000 + FFFF

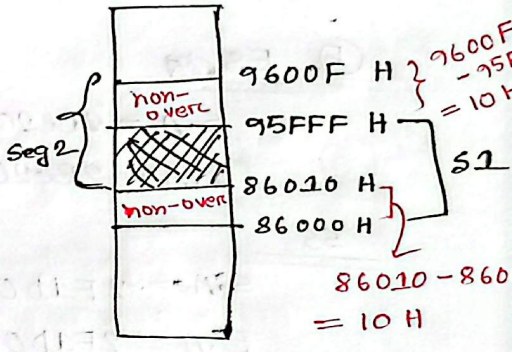
= 95FFF H

Segment 2 :

S.A = 86010 H

E.A = 86010 + FFFF

= 9600F H



বের করতে পারবো না

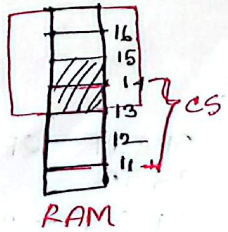
∴ total non-overlapping = 10 H + 10 H

= 20 H.

~~Non~~ overlapping slot বের করা :

overlapping : 95FFF – 86010 +1 → always করা লাগবে।

= FFF0



valid হবে।

previous trimester question.

Spring 26

3. (a) DS

offset = 2BCEC − 1E5B * 10
= D7BC (valid)

ES = 2BCEC − 2BC2 ×10
= 00CC H (valid).

SS

offset = 2BCEC − 2F1D * 10
= FFFF CB1C (invalid)

(b) ES

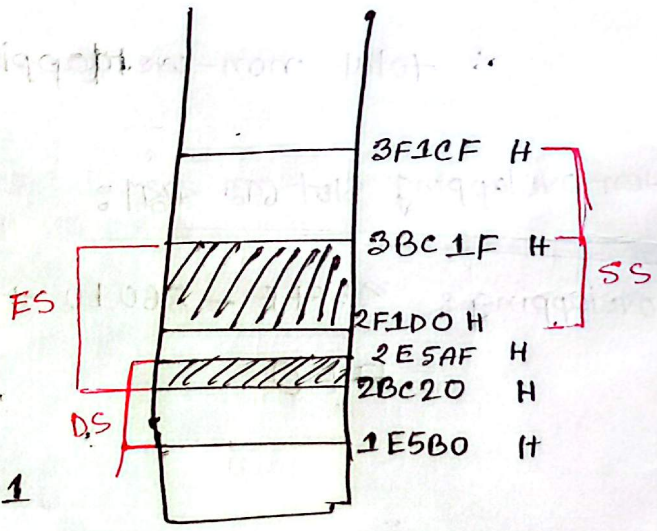
S.A = 2BC20
E.A = 2BC20 + FFFF = 3BC1F H

SS

S.A = 2F1D0
E.A = 2F1D0 + FFFF = 3F1CF H

DS

S.A = 1E5B0
E.A = 1E5B0 + FFFF
= 2E5AF H.



Overlap

(ES, SS) = 3BC1F − 2F1D0 + 1
= CA50

(ES, DS) = 2E5AF − 2BC20 + 1 = 2990

CamScanner

PA – offset = Segment * 10 H

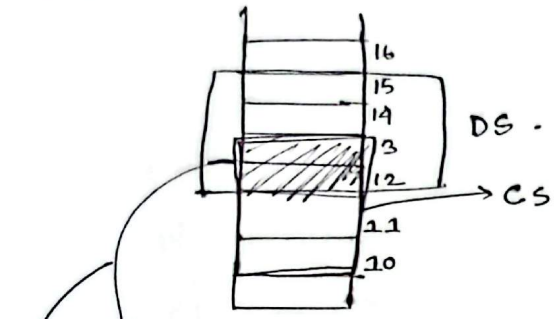
S.A.

S.A/10H = segment.

~~Fall 25~~ segment বের করতে বলেনি,

চিন্তা:

কেন +1 যোগ করি overlapping এ:



overlapping = 13 – 12 + 1

= 1 + 1

= 2

বিয়োগ করার পর
1 টা slot আছে কিন্তু
ভেতরে দুইটা slot তাই
1 যোগ।

overlapping এ last overlapping থেকে first overlapping বিয়োগ দিলেই আমরা আগেই overlapping এর সংখ্যা বাদ দিয়ে দিচ্ছি। তাই answer ঠিক করার জন্য +1 যোগ করি।

Fall 25

CS = 35E9

DS = 2BC1

~~SS = 1F5B~~ ×

ES = 2F1D

SP = 3344 ×

IP = 2211

SI = 11AA

extra value.

CS

S.A = 35E90 H

E.A = 45E8F H

DS

S.A = 2BC10 H

E.A = 3BC0F H

ES

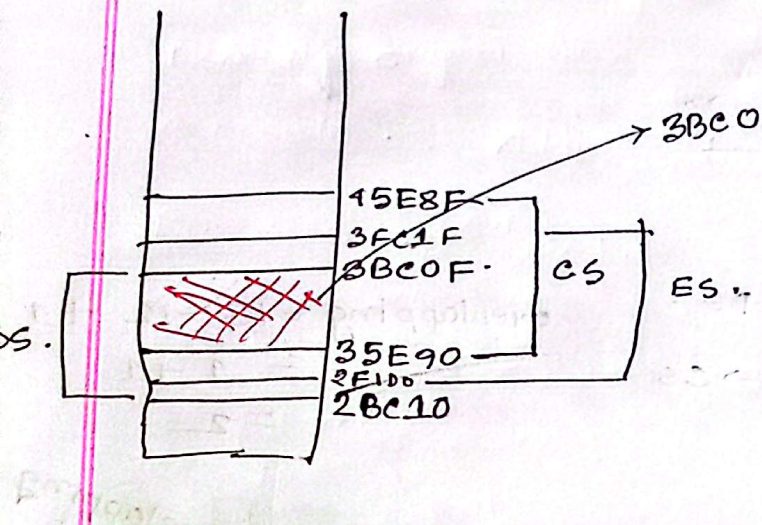
S.A = 2F1D0 H

E.A = 3F1CF H

CamScanner

9/08/2026.
Sunday :)

Topic: Bios and memory interfacing.



Summer 25.

(3) (a) Segment * 10H = P.A - offset.

Stack segment,

P.A = SS * 10H + SP

= 7CA2 * 10 + 3344

= 7FD64

Stack segment এর জন্য P.A বের করতে segment SS and offset SP/BP.

Segment * 10 = 7FD64 - B294

S.A = 74AD0 → valid

∴ Segment = 74AD0 / 10

= 74AD0.

12/08/2026.
Wednesday.

Topic: 8086 and memory interfacing.

Memory organization:-

connections: Address lines, Data, Enable and control.

Ram size = $2^{A \cdot B} \times D \cdot B$ → যতো সিগন্যাল থাকবে num of locations.

Enable lines:-

CS
chip select

CS = 0 [RAM off]
CS = 1 [RAM ON]

control line: pin = $\overline{WE}$
write, enable

$\overline{WE}$ = 0 (write)
$\overline{WE}$ = 1 (Read)

Active high (A)
→ 1 (Active)
→ 0 (Inactive)

Active Low ($\overline{A}$)
→ 0 (Active)
→ 1 (Inactive)

Math 1: given, Data Bus = 16-bit
address Bus = 20-bit.

Ram size = $2^{A \cdot D} \times D \cdot B$,

$= 2^{32} \times 16$ bit

$= \frac{2^{32} \times 16}{8}$ Byte

$= \frac{2^{32} \times 16}{8 \times 2^{30}}$ GB

$= 8$ GB

(2) Data Bus = 32 bit
Address Bus = 16 bit.

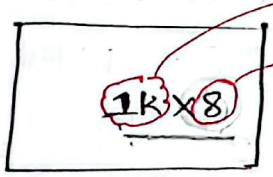
∴ RAM size = $2^{16} \times 32$ bit

$= \frac{2^{16} \times 32}{8}$ Byte

$= \frac{2^{16} \times 32}{8 \times 2^{10}}$ KB

$= 256$ KB.

RAM Block Diagram.



→ number of memory locations = 1K = 1024

→ Data Bus = 8 bit

A.B = n bit

$\Rightarrow 2^n = 1024$

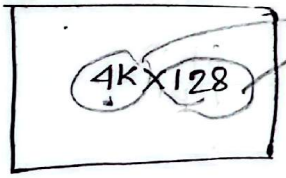
$\Rightarrow n = \log_2(1024)$

$\Rightarrow n = 10$.

1 bit = 0, 1

$= 2^n$

(1)



D.B = ?

A.B = ?

Ram size = ?

$2^n = 4K$

∵ D.B = 128 bit.

∴ $2^n = 4K$

$\Rightarrow 2^n = 4 \times 1K$

$\Rightarrow 2^n = 4 \times 1024$

$\Rightarrow n = \log_2(4 \times 1024)$

$\Rightarrow n = 12$.

∵ Ram size $= 2^{12} \times 128$ bit

**

Memory Decoding (6 mark)

Linear decoding

microprocessor, A·B = 16 bit
D·B = 8 bit

RAM, A·B = 10 bit
D·B = 8 bit

$2^{A \cdot B} \times D \cdot B$

Step 1: RAM size calculate করতে হবে, $\rightarrow 2^{A \cdot B} \times D \cdot B$

$$\text{RAM-size} = (2^{10} \times 8)\ \text{bit}$$

$$= \frac{2^{10} \times 8}{8}\ \text{Byte}$$

$$= \frac{2^{10} \times 8}{2^{10} \times 8}\ \text{KB}$$

$$= 1\ \text{KB}$$

Step 2: num. of RAM needed বের করবো,

$$\therefore \text{No of RAMs Needed} = \frac{\text{interfacing size}}{\text{size of each RAM}}$$

$$= \frac{6\text{KB}}{1\text{KB}}$$

$$= 6$$

$\frac{6}{4} = 1.5$
নিবো $= 2^2$

Step 3: আবার upore draw করবো micro. processor.

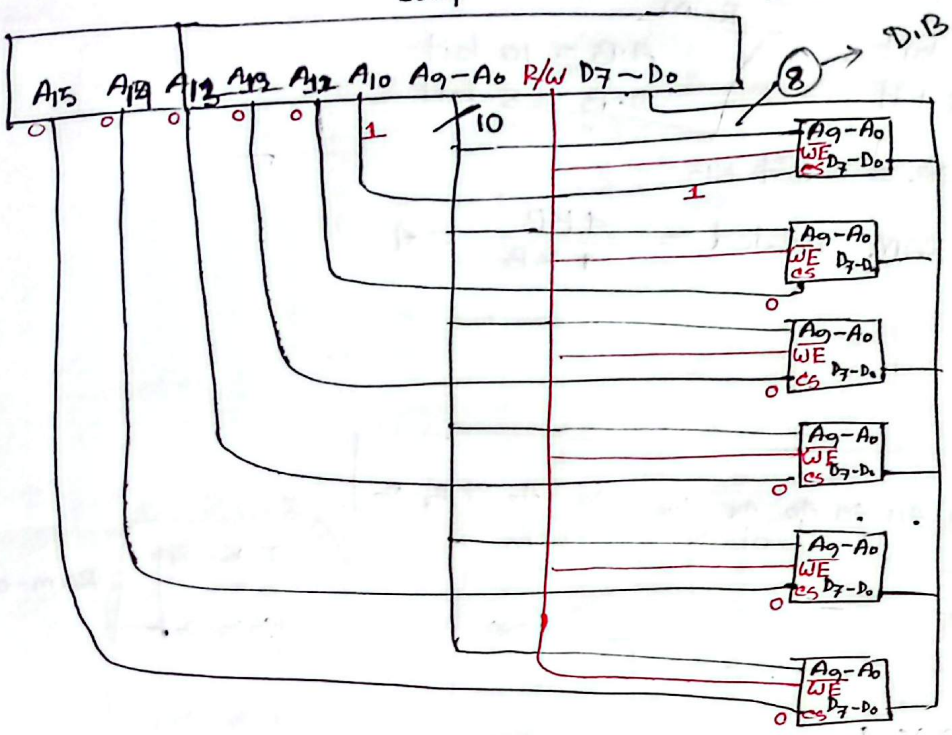
→ first এ Data-bus এর connection দিবো
→ then A·B এর connection.
RAM এর যত bit লাগবে তত bit নিবো
→ control line connection দিবো (WE এর সাথে R/W)

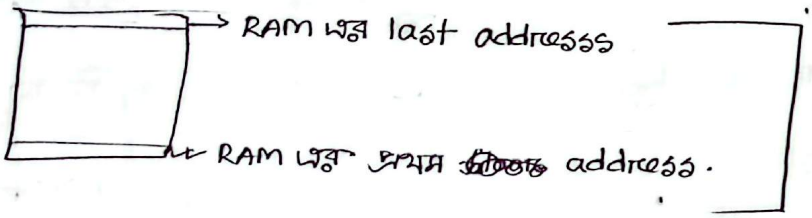
CamScanner

RAM-এর First location বের করার জন্য → linear decoding

RAM এর A·B যেই কয় bit সেই কয়টি = 0

Last location এর A·B যেই কয় bit সেই কয়টি = 1 হবে।

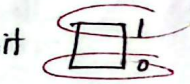




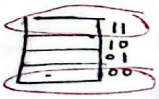
প্রত্যেক RAM এর জন্য বের করবো

A_{15}	A_{14}	A_{13}	A_{12}	A_{11}	A_{10}	$A_9 \cdots\cdots A_0$	
0	0	0	0	0	1	0 ····· 0	RAM-0
0	0	0	0	0	1	1 ····· 1	

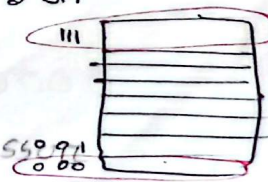
A·B = 1 bit



A·B = 2 bit



A·B = 3 bit



* এক একটা location এর address পাঠানোর মাধ্যম microprocessor এর

* Microprocessor এর A·B = 16 bit. তাহলে processor-এ

→ ৫ এর connection খুলিয়া আমরা processor এর বাকি Address pin দুইবে। processor এর address pin 10 টা use করেছি। এখনও 6 টা বাকি। এই 6 টা থেকে ৫ এর connection খুলিয়া যাবে।

প্রত্যেক Location এর জন্য 16 bit এর address generate করুন।

Full decoding.

µ.p.
A.B = 16 bit
D.B = 8 bit

RAM
A.B = 10 bit
D.B = 8 bit.

RAM size = 1 KB

No. of RAM needed = $\frac{4\,KB}{1\,KB} = 4$

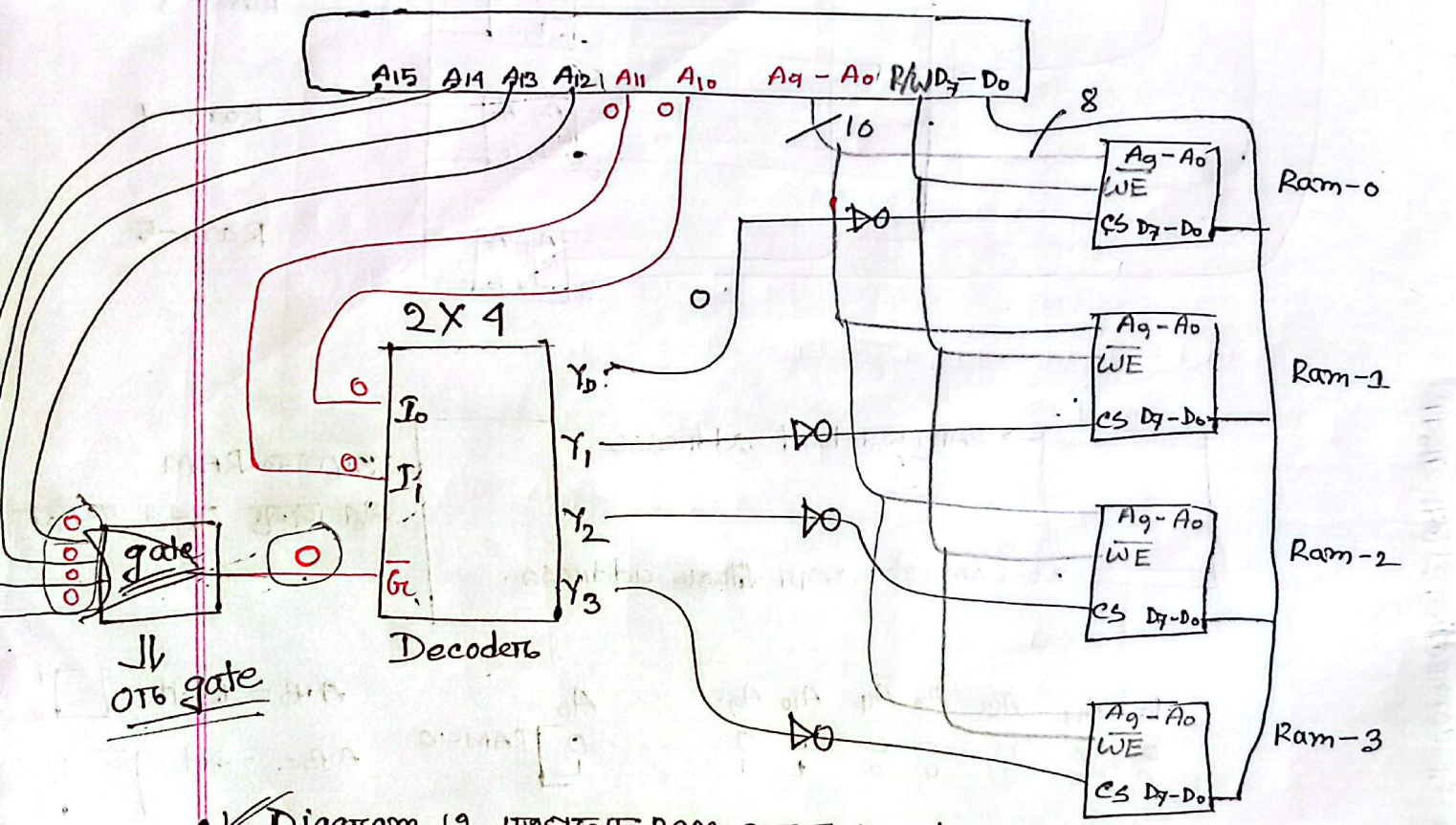
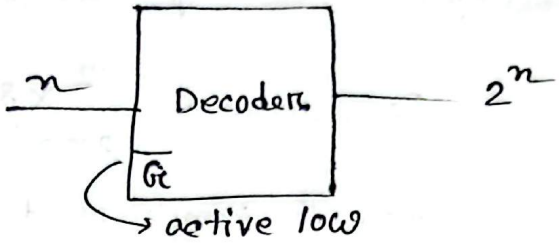


Diagram এ যতগুলো RAM থাকবে Decoder এ minimum ততটা output থাকবে। তাহলে input 2 টা।

0000 H

input = n
output = 2^n



16 August 2026.
Sunday.

① Problem 1 (Linear)

Part-2.

Step 1: Ram size = $2^{12} \times 8$ bit

$= \frac{2^{12} \times 8}{8 \times 2^{10}}$ KB

$= 4$ KB

μ-p
A.B = 16 bit
D.B = 8 bit.

Ram
A.B = 12 bit
D.B = 8 bit.

Step 2: no. of RAMs needed = $\left[\frac{9KB}{4KB}\right] = 3$.

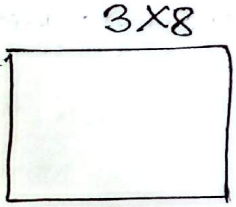
~~#~~ Design করা লাগবে।

② Full Decoding : Decoder use করবো

RAM size = $\frac{2^{10} \times 8}{2^{10} \times 8}$ KB = 1KB

no. of Rams needed = $\left[\frac{7KB}{1KB}\right]$

= 7.



μ-
A.B = 16 bit

RAM A.B = 10 bit.

CamScanner

Important ***

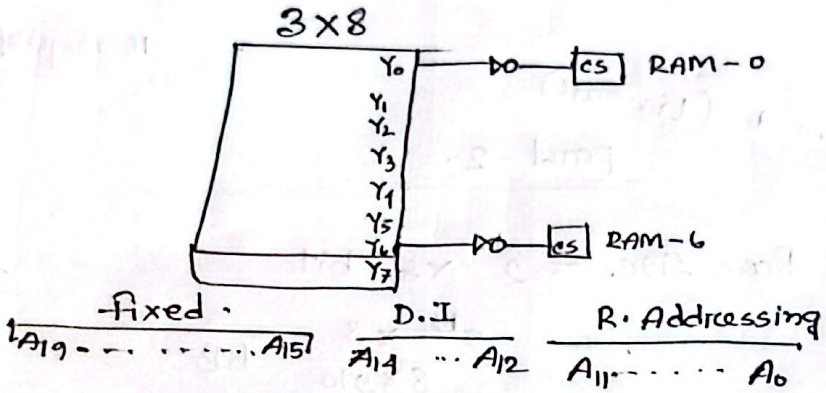
1st ram এর first add. ↓ last ram এর last add. ↓

Slide-16(page) memory range 50000H – 56FFFH

1(b)

$$\text{Ram size} = \frac{2^{12} \times 8}{8 \times 2^{10}} = 4\,KB$$

$$\text{No. of Rams Needed} = \left[\frac{28}{4}\right] = 7.$$



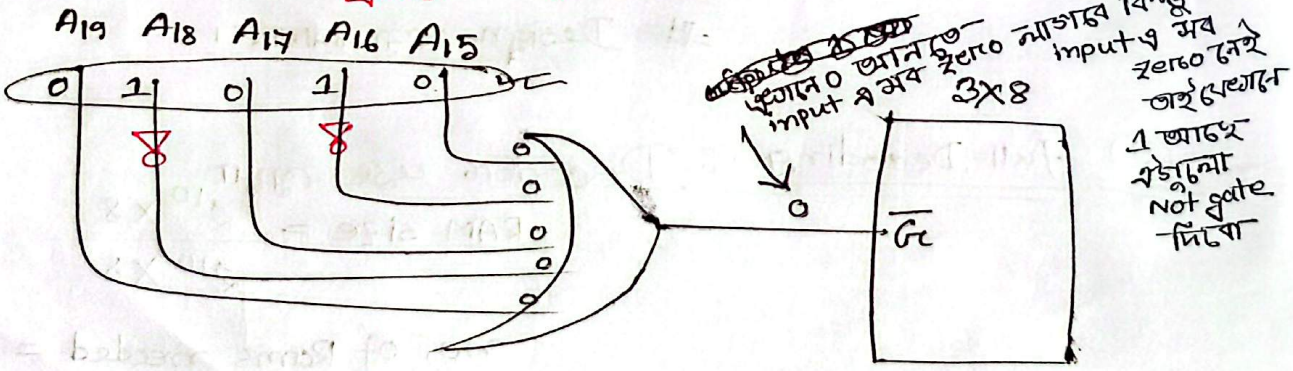
Decoder in

50000H = 0101 0000 0000 0000 0000

56FFFH = 0101 0110 1111 1111 1111

RAM addressing

fixed এখানে change



Question এ যেই range দেয়া থাকবে তাদেরকে binary তে নিবো.

Spring 21

second ram এর starting address = 2D800 H

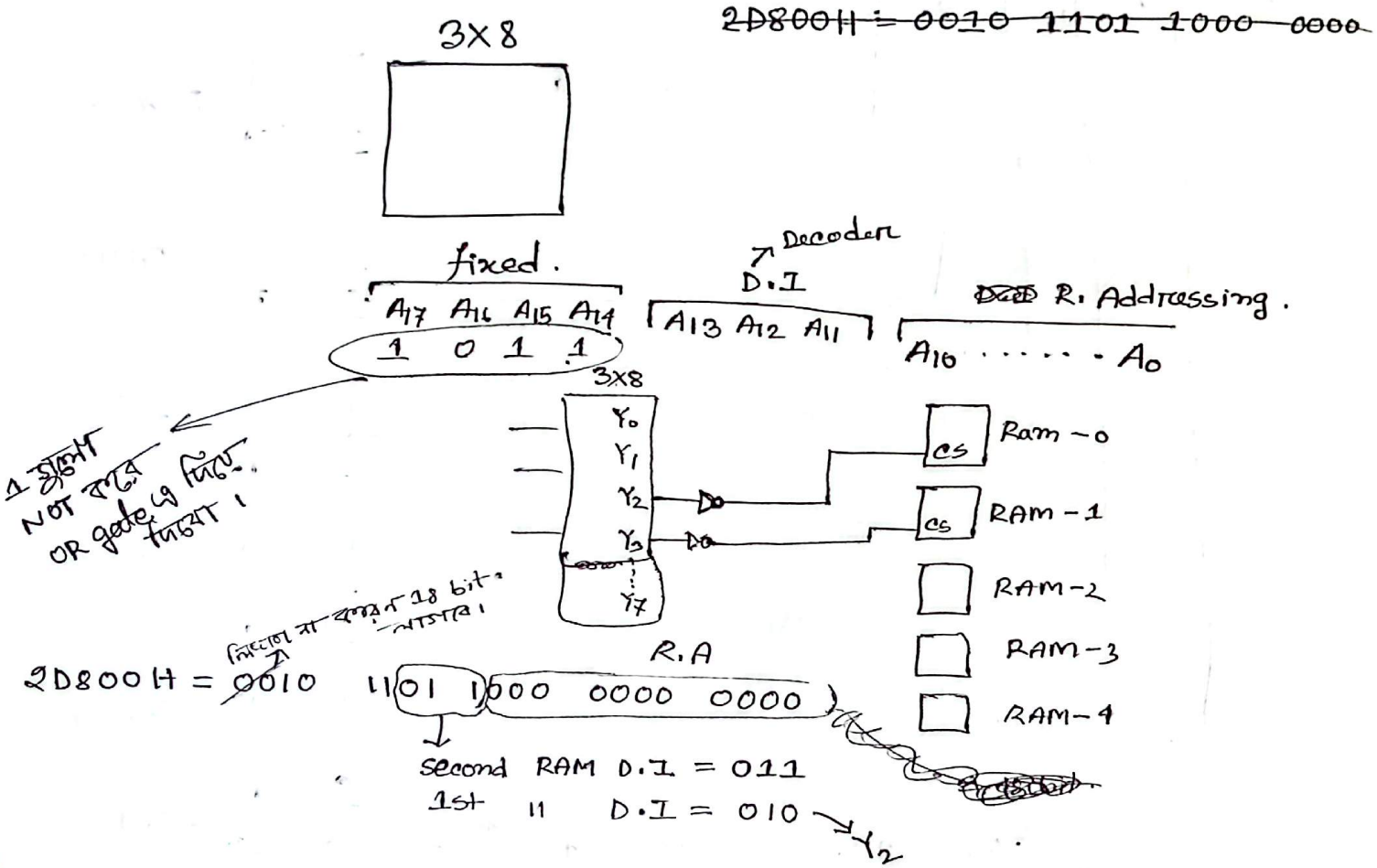
4.(a)

Ram size $= 2^{11} \times 16$ bit

$= \frac{2^{11} \times 16}{8 \times 2^{10}}$ KB $= 4$KB

No. of Ram needed $= \left\lceil \frac{18\text{ KB}}{4\text{KB}} \right\rceil = 5$

~~2D800H = 0010 1101 1000 0000~~



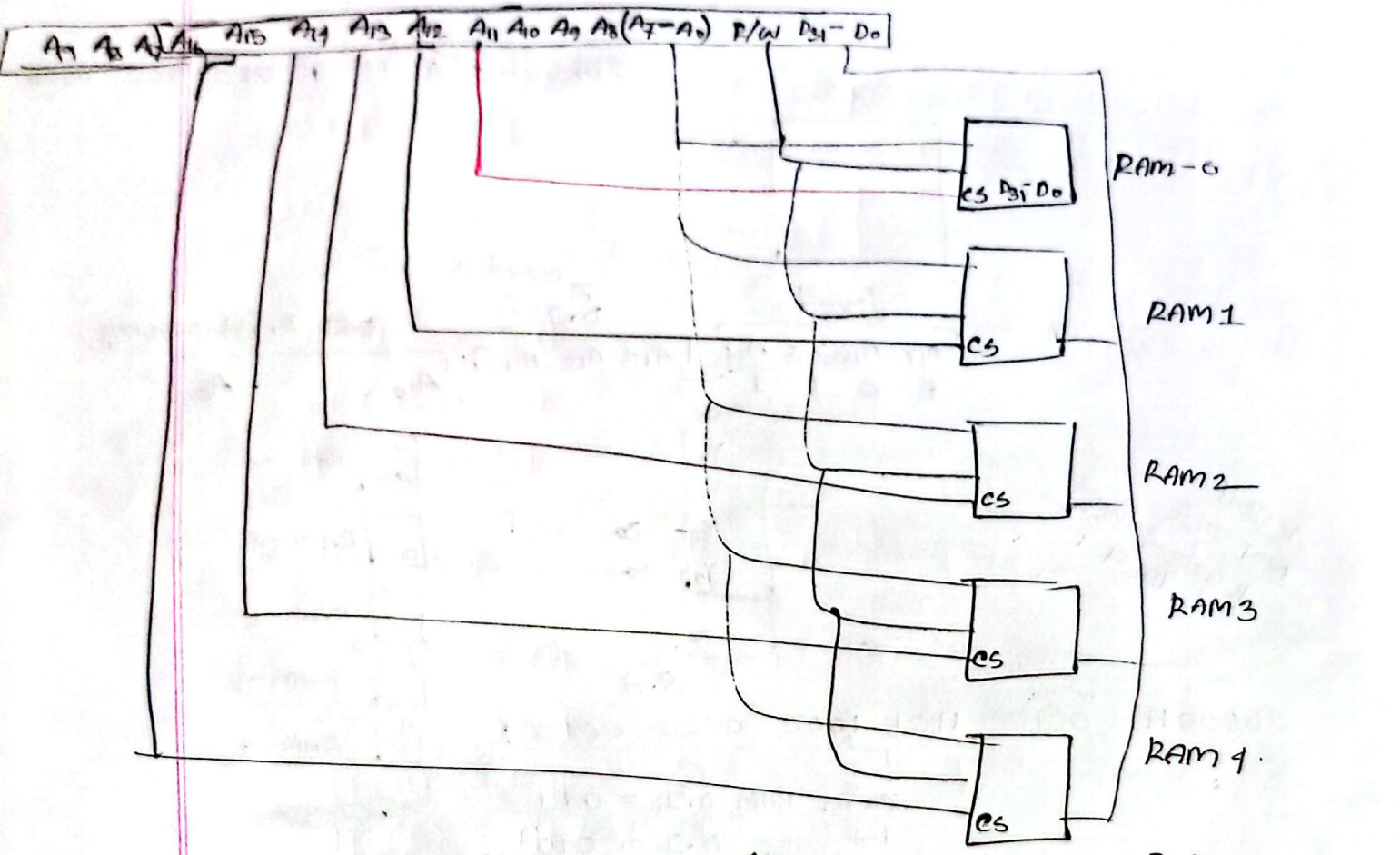
last RAM D.I = 111 → Question এ বলে দিবে design এ last RAM এর CS টা Y_7 এর সাথে connect হবে।

Fall 25. (linear)

RAM size. $= \frac{2^8 \times 32}{8 \times 2^{10}} = 1\ KB$.

No. of Rams needed $= \left\lceil \frac{5KB}{1KB} \right\rceil = 5$.

the first RAM starts = 00800H



00800H = 0000 0000 1000 0000 → R.A

A11 A10 A9 A8

- প্রথম RAM এর জন্য A8 এ 1 আসার কথা (CS connection) কিন্তু 1 আছে A11. তাই A11 থাকে RAM-0 এর CS এ যাবে

- normally zero থাকে। এখনও আছে তাই change হয়নি।

CamScanner

Summer 25

RAM size $= \frac{2^{10} \times 6}{8 \times 2^{10}} = 2KB$; No. of RAMs needed $= \left[\frac{10KB}{2KB}\right] = 5.$

Acc00 = 1010 110 0 110 00 0000 0000 → R.A

(fixed) (D.I) (R.A)

1st RAM DI = 0110

fixed: $A_{19}\ A_{18}\ A_{17}\ A_{16}\ A_{15}\ A_{14}\ A_{13}$ | D.I: $A_{12}\ A_{11}\ A_{10}$ | R.A: $A_9 \dots A_0$

Y3 এর সাথে connect হবে- প্রথম RAM এর CS.

Topic : 8086 interrupt

total interrupt = 256 (0 - 255)

INT 00

INT FF

Formula :

Physical Address of IP(Low) = INT number * 4 H.

Math 1

INT C H → Hexa.

P.A IP(L) = C X 4

= 30H IP(L)

= 31 IP(H)

= 32 H → CS(L)

= 33 H → CS(H)

Math 2

INT EH

= E H * 4H

= 38H IP(L)

= 39 IP(H) ✗

= 40 CS(L)

3A

41 = 3B

3A, 3B → এভাবে হবে।

19/08/2026.
Wednesday.

Interrupt (0-255) এর বেশি আসলে invalid.

NMI → interrupt আসলে execute করতে হবে-ই,

Register → 16 bit

RAM এর প্রত্যেকটা slot → 8 bit.

* Multiple interrupt আসলে যার priority বেশি তাকে execute করবো। priority depend করে interrupt numbers এর উপর।

INT numbers ↓	Priority ↑

(*) CS(L) = 6H
IP(H) = 5H
IP(L) = 4H
-2

P.A of IP(L) = INT num * 4H

∴ INT numbers = $\frac{4H}{4H} = 1$

(*) CS(H) → 7H
CS(L) → 6H
IP(H) → 5H
IP(L) = 4H
-3

* একটা interrupt আসলে processor কীভাবে কাজ করে?

→ i) flag register stack এ push.
ii) IF, TF : zero করবো
iii) CS এর value push
iv) IP " " "
v) then interrupt execute করবে।

(Fall 25)

(a) Interrupt = 80 ঢ
(0 - 79)
→ Hexadecimal এ নিবো। (4F H) → (0 - 4F)

∴ P.A of IP(L) = INT number * 4 H
= 4F * 4H
= 13C H → IP(L)
13D H → IP(H)
13E H → CS(L)
13F H → CS(H)

(b) For Denji's :

IP(L) = $\frac{00138\ H}{4\ H}$ = 4E H [0 - 4F এর মধ্যে]

Aki's :

CS(H) = 00193 H
CS(L) =
IP(H) =
IP(L) = 00190 H (-3)

$\frac{00190\ H}{4}$

= 64 H → 100 → Decimal এ → [0 - 79 এর মধ্যে]
↓
invalid

(c) 4E H = 78
64 H = 100
→ interrupt number যার কম তার priority বেশি। এখানে interrupt num হলো 4EH যার 80 আগে execute হবে।

Summer 25

5 @ একটা interrupt এর জন্য total জায়গা লাগে 32 bit / 4 Byte.

∴ 4B জায়গা লাগে 1 interrupt এর জন্য

∴ 1B " " $\frac{1}{4}$ " "

∴ 2×2^{10} B ∴ $\frac{1 \times 2 \times 2^{10}}{4}$ " "

$= 512$ " "

$= (0-511)$ ⟶ decimal এ আছে।

↓

1FF H

∴ P.A of IP (L) = INT num * 4 H

$= 1FF * 4$ H

$= 7FCH \rightarrow IP(L)$

$= 7FDH \rightarrow IP(H)$

Ans $= 7FE\ H \rightarrow CS(L)$

$= 0 \rightarrow CS(H)$

⑥

2003
1A20
F202

Stack

IF = 0

TF = 0

P.A = CS * 10H + IP

⇒ CS * 10 = P.A − IP

⇒ CS * 10 = 1C203 − 2003

⇒ CS = $\frac{1A200}{10}$ → valid.

⇒ CS = 1A20

Recap

RAM A·B = 8 bit

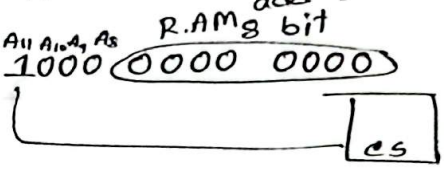
Linear decoding.

μ-P A·B = 16 bit.

RAM সাজাবে = 4 টি

first RAM এর first address = 0800H.

= 0000 1000 0000 0000



RAM A·B = 8 bit.

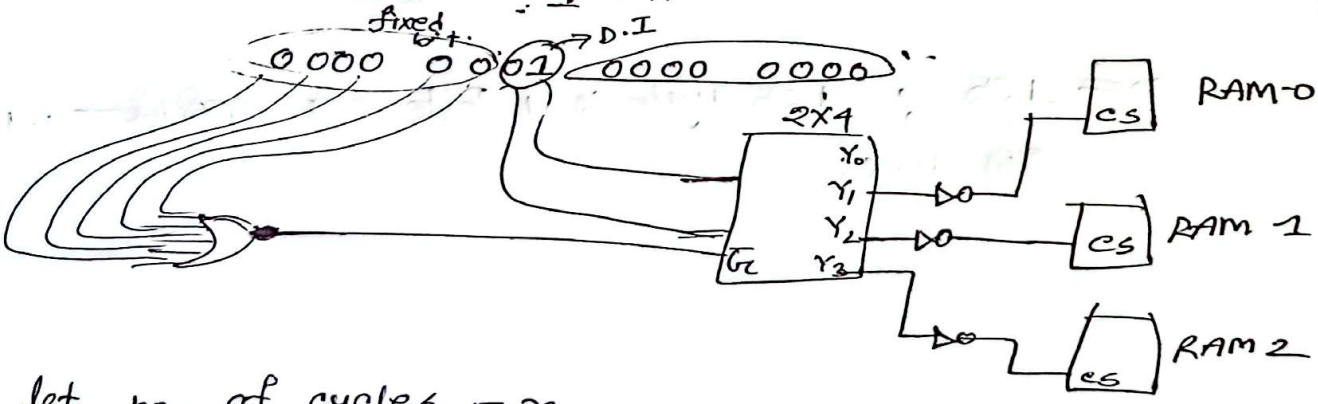
full decoding.

RAM সাজাবে ৪ টি

μ·P

A·B = 16 bit

1st RAM 1st add = 0100H



Ans: (4)

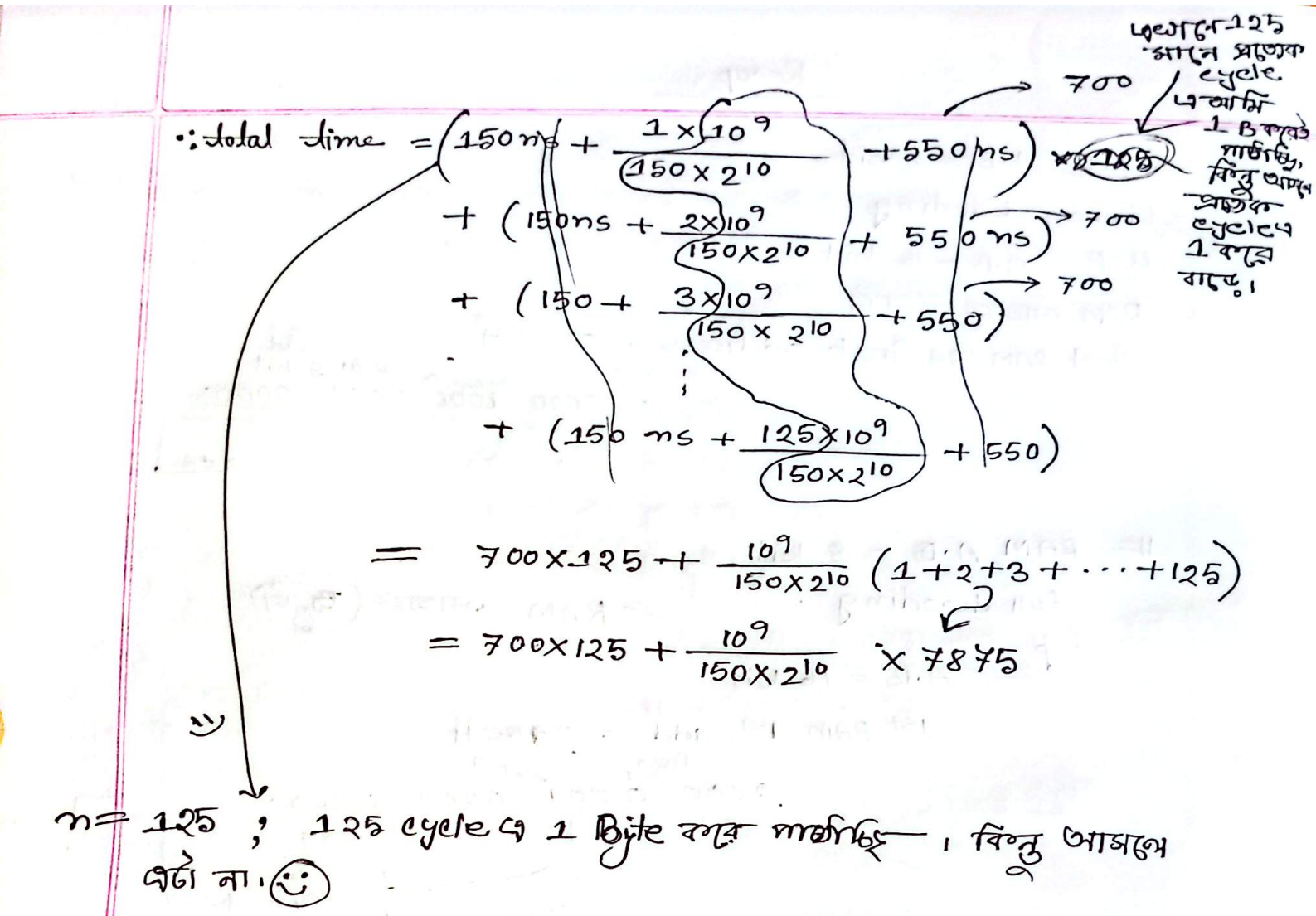
let, no. of cycles = n

$1+2+3+4+\cdots+n = 7875$

$\Rightarrow \frac{n(n+1)}{2} = 7875$

$\Rightarrow n^2+n-2\times7875=0$

$n = 125$



$\therefore$ $n = 125$; 125 cycle এ 1 Byte করে পাঠাচ্ছি। কিন্তু আসলে পাঠাচ্ছি না। ☺